

Image Processing Techniques for Lobster Grading System

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Abstract— Lobster is a fisheries product with a high selling value; fishermen in coastal areas sell their harvest on traditional marketplaces and export it internationally. The high level of fishing pressure brought on by the rising demand has decreased the potential output from the fishery and the associated economic benefits. The BFAR and DA have put in place practical methods to aid in lobster population recovery. The illegality of capturing, transporting, having, dealing, and selling gravid spiny lobsters and juvenile spiny lobsters from the wild that are less than the maturity Carapace Length (CL) is one of the probations underlined in the fisher administration. The study focuses on automating the manual way of measuring the lobster size using the gauge. It employs a visual computing technology that uses a camera to capture images of a lobster and automatically determine its corresponding measurement in such a way that the grading procedure may be carried out immediately without affecting the form and physicality of the lobster, which is especially crucial when the lobster is still alive. Evaluation of the correctness of the grading resulted in an 83.33% accuracy.

Keywords— *image processing, visual computing, lobster grading, segmentation*

I. INTRODUCTION

One of the richest and most diversified coastal ecosystems on the planet is found in the Philippines [1]. In both the domestic and international markets, lobster is one of the most sought-after high-value commodities. In addition to being able to satisfy domestic demand, the Philippines exports lobster to a large number of nations in Asia, Australia, and North America. Most of the world's lobster markets are located in Asia, mainly in China, Hong Kong, Japan, Malaysia, Singapore, Vietnam, the Republic of Korea, and Thailand. These exports enable the development of markets in addition to providing a return on investment for the country [2].

The Philippines is a major exporter of lobster seeds. As stated in the Philippines Lobster Market Insights, a total of 245 and 1,086 tons of lobster were exported by the Philippines in 2020 and 2019, respectively. The Philippines' lobster demand decreased in 2019 alone, varying by -27.114% less than it did in 2018. The country earned US\$4.89 million from exporting lobster in 2019, a -60.79% decline from 2017. In terms of lobster exports, the nation comes in at number 45.

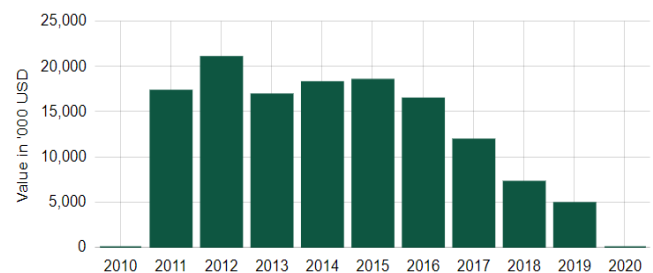


Fig. 1. Philippines lobster export values

A high amount of fishing pressure is being caused by the indiscriminate catching, selling, and trading of the species due to the rising demand. This further had the effect of lowering the possible yield from the fishery and the prospective economic gains from the stock [3]. Because of the enormous economic demand for lobsters and their high price, small-scale fishermen are keen to catch them. As a result of this commercial exploitation, the number of adult and fry stages being captured is decreasing [4]. In 2016, lobster production significantly decreased, from 71.88 metric tons in 2008 to 6.10 metric tons [5].

The government is trying to address the continuous decline of lobster export values. The Philippines through the Bureau of Fisheries and Aquatic Resources (BFAR) has issued the Philippine National Standard (PNS) – Code of Practice for harvesting and processing of lobsters as one of its standardization processes of adopting relevant international standards. Fishing Administrative Order No. 265 Series 2020 was also issued by the Department of Agriculture to regulate the catching, possession, transporting, selling, trading, and exports of lobster.

Workable mechanisms to help the recovery of lobster populations remain a formidable challenge. The illegality of capturing, transporting, having, dealing, and selling gravid spiny lobsters and juvenile spiny lobsters from the wild that are less than the maturity. Carapace Length is one of the probations underlined in the fisher administration (CL) [3]. The size of a lobster is assessed after it has been captured. The lobster's length is calculated by measuring from the rear of the eye to the back of the shell or carapace. The legal minimum is 3 1/4 inches, while the maximum is 5 inches. Anything that is either lower or higher than the minimum must be returned right away to water where it was found. Disregarding these standards entails high cost and serious penalties for every violation [6][7].

To provide assistance and additional insights on lobster harvesting, a tool is needed for grading the lobster base on their measurement. The study focuses on automating the

manual way of measuring the lobster size using a gauge. It employs a visual computing technology that uses a camera to capture images of a lobster and automatically calculate its corresponding measurement in such a way that the grading process can be carried out directly without affecting the lobster's form and physicality, which is especially crucial when sold alive. This computer vision grading approach is thought to be particularly effective in enhancing the export quality and lobster value.

II. RELATED STUDIES

Computer vision is a fascinating research field that investigates ways to help computers sense, interpret, and comprehend visual data such as photos and movies. Likewise, it includes image processing as a subset. Moreover, image processing can be used to enhance an image's quality, eliminate unwanted objects from an image, or even produce new images from scratch. As this is a wide and complicated discipline with several methods and approaches that may be utilized to produce various outcomes. Some studies focus on the grading system, for which numerous researchers utilized various methodologies.

In an aquaculture application, Vo et al. [8] used the Convolutional Neural Network (CNN) in combination with image processing approaches to allow an automatic grading solution in the Southern Rock Lobster supply chain in Australia and China. The creation of a low-cost biometric identification technology for tracking lobsters from catch to table. The developed method is capable of utilizing low-cost mobile cameras [9]. Their research applied CNN to improve prior study on grading lobster by building a dependable and adaptable traceability mechanism to satisfy varied supply chain scenarios. Using a pre-trained Mask-RCNN model, areas of interest from lobster pictures were extracted. The model's deep learning capability allows the carapace portions of lobster photos to be automatically separated for determining grading criteria such as size, weight, and color. Another method employed by Vo et al. [10] was a Siamese model paired with a contrastive loss function to distinguish between different lobsters based on pictures of their carapaces. These places are said to have distinct traits generated by colors and spiky patterns.

Hu et al. [11] employ computer vision to classify fish species using color, texture, and a multi-class support vector machine. The smartphone was used to capture and send fish photographs, and the approach used included the following phases. Original pictures were used to create color and texture sub-images of fish skin. Color features, statistical texture features, and wavelet-based texture features were extracted from the color and texture sub-images, and six groups of feature vectors were created. To discover the optimum group for classification in the feature selection method, the LIBSVM program was validated using leave-one-out cross validation. For classification, two multi-class support vector machines based on a one-against-one method were built. The Bior4.4 wavelet filter in HSV color space obtained more accuracy than the other feature groups, according to the feature selection findings.

Johansyah et al. [12] used machine vision technology as method for grading lobster to increase the value of selling it. The gray-level-co-occurrence matrix (GLCM) method used to quantify texture.

Moreover, fuzzy logic approach is also used in an image processing. Fuzzy image processing encompasses all techniques to understanding, representing, and processing pictures, their segments, and characteristics as fuzzy sets. The representation and processing are determined by the Fuzzy approach used and the problem to be solved. The outcomes of fuzzy logic applications can be used in the feature extraction and object recognition stages of image processing, as well as future computer vision. The paper of Razak et al. [13] designed and implemented approaches and algorithms for determining the quality of local mango production in Perlis using digital fuzzy image processing, content-based analysis, and statistical analysis. The primary contribution of this work is the creation and construction of an efficient algorithm for recognizing and sorting mangoes with greater than 80% accuracy in grading when compared to human expert sorting.

OpenCV is likely the most commonly used computer vision library, with hundreds of ready-to-use image and vision functions that are utilized in both academia and industry. As cameras become more affordable and image features become more popular, the number of apps that use OpenCV expands considerably, both for desktop and mobile platforms [14].

III. FRAMEWORK OF THE STUDY

The framework shown in Fig. 2 consists of a data collection of sample images of lobsters using digital or cellphone cameras. These images were categorized in accordance with the grade determined by human expert assessment and the standard established by the BFAR. Each subset comprises various grades that signify the classification or grading of the lobster. The obtained images were then subjected to a number of image processing techniques to prepare them for information extraction.

Additionally, the accuracy of the system's classification of the image as belonging to the right category was calculated as a fundamental evaluation metric.



Fig. 2. Research conceptual framework.

IV. METHODOLOGY

Fig. 3 describes the processes covered in this study that utilized the image processing and lobster grading techniques.

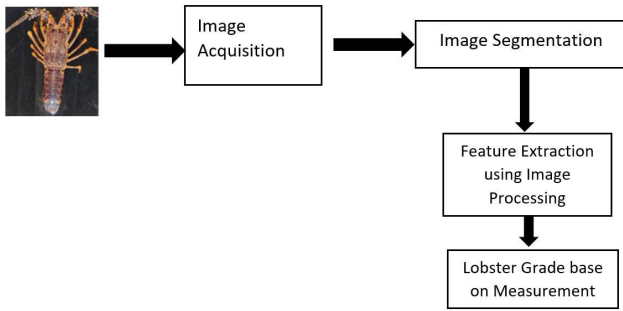


Fig. 3. Block diagram of the study.

A. Image Acquisition

Image samples of lobsters were considered as dataset for this study. Each image was manually classified by the human expert for its appropriate grading. Table I shows the different lobster grading.

TABLE I. LOBSTER GRADING CODE

Grading Code Base on Measurement	Carapace Length	Description
A	Greater than 5 inches	For Breeding (Not for harvest)
B	Greater than or equal to 3.5 inches and less than or equal to 5 inches	For Harvest/Market
C	Lesser than 3.5 inches	Not for Harvest

For the dataset of this study, each lobster was placed on top of a plain surface and an image of it was captured using a cellphone camera. Excessive was avoided when taking the image since it would interfere with the outstanding image processing output. Additionally, the camera distance was taken into consideration because it is crucial in determining the size. In this instance, a distance of 40cm from the lobster carapace was applied. Moreover, the original picture resolution was configured to 2400x1080, which provides a high degree of detail for classification process.

B. Image Pre-processing

In this phase, several techniques were used to pre-process image. These include, image resizing, median filtering to enhance image quality and grayscaling to reduce computational requirement.

C. Image Segmentation

Researchers performed an edge detection as a fundamental step in image processing in order to trace the edges of the image to get better results. This method was also used to determine the boundaries of objects within an image.

Then, the threshold-based image segmentation approach was applied to split the region of interest, which is the lobster, and the background. The Otsu technique was used to determine the optimal threshold value. Its main objective is

to divide pixels in an image into two segments: the object (foreground) and the background.

D. Feature Extraction and Classification

The region of interest was obtained and was subjected for feature extraction to determine the measurement (length and width) of the carapace. The calculated values specifically the length, were considered as basis for automatic lobster grading.

In this phase, the image was extracted and compared with the relevant input data. This method extracted only the image of the lobster. The said image was relevant to the input data that will be extracted.

Lobster grading is categorized into three components namely A, B and C based on the measurement. Grade A implies that lobster should not be harvested since they are intended for breeding. Grade B designates that lobsters should be harvested for marketing purposes. Whereas, grade C means that lobsters are not considered for harvesting.

V. RESULTS AND DISCUSSIONS

To carry out the lobster grading process, the input image was subjected to image processing using OpenCV to detect and extract lobster and the corresponding x and y axis of the object boundaries.

As demonstrated in Fig. 4, the region containing the lobster has been recognized. However, it covers the entire body of the lobster and does not precisely determine the carapace part of the lobster. The focus is on segmentation, with visual computing used to separate carapace regions from lobster photos for grading and analysis. The height of the region of interest is then calculated by manipulating pixels.

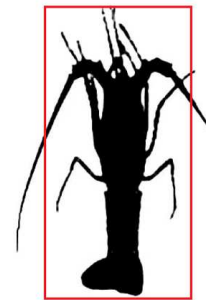


Fig. 4. Detected lobster from the loaded image

After performing some adjustments on the segmentation process, the output now satisfies the required region of interest for which features necessary for lobster grade grading are to be extracted (Fig. 5).

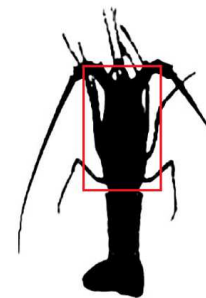


Fig. 5. Detected carapace from the loaded image.

Fig. 6 and Fig. 7 show the user interface of the lobster grading system. The upper leftmost section of the application featured a display of the camera and served as a panel for the input image. After clicking the “Start” button, output of individual processes will be indicated in the bottom part of the window. These include grayscaling, edge detection and feature extraction (measurement of carapace part of the lobster). Finally, based on calculations made on the extracted feature value, the corresponding lobster grading or classification will be displayed.

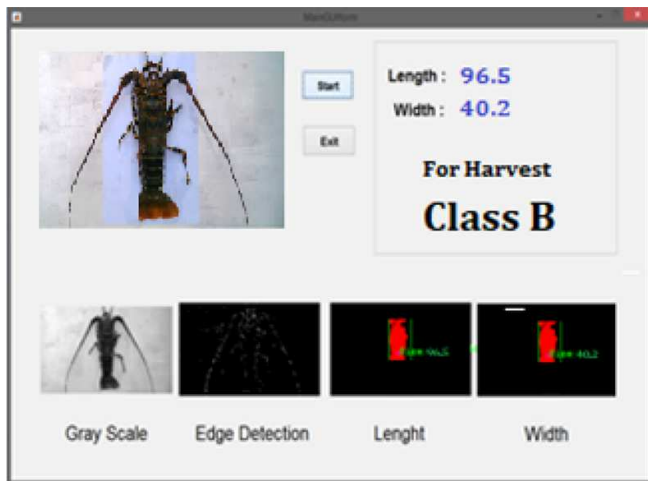


Fig. 6. Sample Grade B classification of lobster.

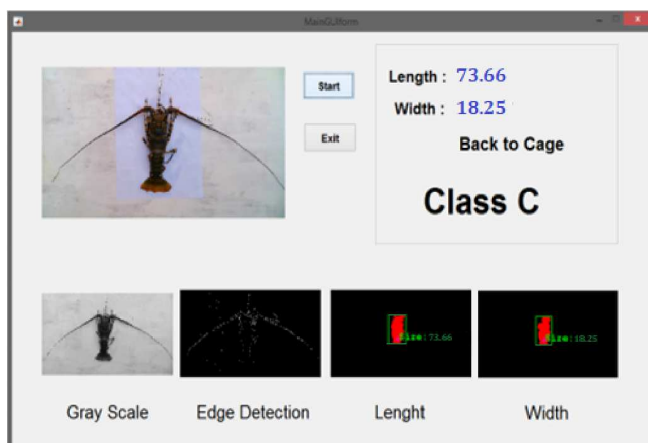


Fig. 7. Sample Grade C classification of lobster.

In testing the performance of the lobster grading system, a total of 30 test images, 10 for each class, were subsequently loaded into the system. Each image's equivalent outcome (lobster grade) was recorded and was then compared to the manual grading done by the human expert beforehand.

Fig. 8 depicts the confusion matrix as the result of the evaluation in which an accuracy rate of 83.33% has been achieved, which indicates a favorable result.

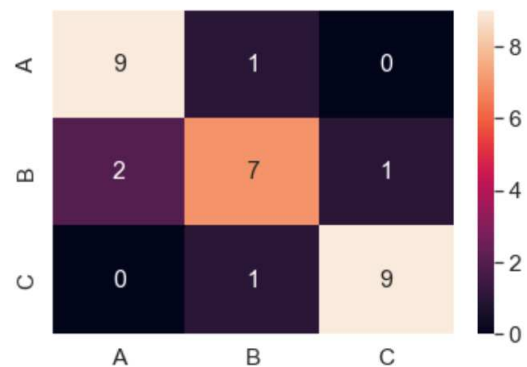


Fig. 8. Confusion matrix

VI. CONCLUSIONS AND RECOMMENDATION

This study was able to produce a system that is capable of capturing the measurement (carapace length) and grading of the image-based lobsters. When the performance of the grading system was evaluated using the test dataset, an accuracy rate of 83.33% was obtained.

The findings of this study signify that the application of computer vision technique would result to a reliable supplemental tool for automated lobster grading. This would help lobsters farmers/growers perform a sensible and timely decisions.

Nevertheless, further studies could be made to improve the overall performance of the lobster grading system. This may involve incorporating additional features or exploring the applicability other image processing techniques or even using a machine learning approach.

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